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Programming Fundamentals PROJECT

Project Report By:

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Zakat Calculator

INTRODUCTION:

The topic of our project is Zakat Calculator. Zakat is the 4th pillar of Islam. It's importance can be recognized by the fact that it is obligate in Islam. Zakat calculator will guide the people that how much zakat they will have to pay according to shariah. As many people do not know how much zakat they will have to pay according to their gold, money and fruits as well. This program will allow the user to enter their current amount they will have and this program will calculate their zakat on that amount.

OBJECTIVES:

The objective of this C++ program is to provide a user-friendly tool for calculating Zakat, adhering to Islamic principles, specifically for Money, Fruits, and Gold. The program aims to streamline the Zakat distribution process by offering a clear interface where the user can choose the type of Zakat they need to calculate and input relevant details. The calculated Zakat amount is then displayed as the output.

Features:

This C++ program calculates Zakat based on user input, with features including:

1. Firstly, the heading of Zakat Calculator will be shown with the help of cout.
2. Secondly, this calculator will say Salam to their users.
3. Then this calculator will ask the following Questions:
 - a. Name:
 - b. Gender:
 - c. Age:
4. The little bit information of distributing Zakat is given in the following paragraph:

Zakat is 2.5% of the wealth of each Muslim in the world. It's paid by every Muslim once a year as ordained by Allah, as a means of ensuring the re-distribution of wealth to those who need it, subhanAllah.

5. Choose your choice from the following options on which you want to calculate the Zakat:
Money, Fruits, Gold, Land, Silver.
6. Calculation of Zakat for money by reading the amount from a file (money.dat) and applying the formula (Money value * 2.5%).
7. Calculation of Zakat for fruits by prompting the user for selling price and expenses, applying the formula (Fruits Zakat = (selling price - expenses) * 2.5%).
8. Calculation of Zakat for gold by asking the user for the amount of gold in grams over 85g, determining if Zakat is applicable, and applying the formula (Gold Zakat = amount of gold * 1g value * 2.5), considering 1g gold worth 10 KD.
9. Calculation of silver and land is also given with their formulas.
10. In the last, this program will take feedback from the user regarding the usage of this program and will say Allah Hafiz to the user.
11. Other than this formulas, if-else and switch statements are used in this program. We can also use if-else-if rather than switch statements.
12. A user-friendly interface that prompts the user to choose the type of Zakat and displays the calculated Zakat amount.

Algorithm:

1. Start
2. Initialize variables for prices and values.
3. Collect personal information (name, gender, age).
4. Provide information about Zakat and its purpose.
5. Use a do-while loop for the user to choose the type of wealth or exit.
6. Switch between different options (money, fruits, land, silver, gold) and calculate Zakat accordingly.
7. Display Zakat values for each type of wealth.
8. Collect feedback and say Allah hafiz.
9. Stop.

Program:

```

1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main()
6  {
7      string name, gender, feedback;
8      int age, choice;
9
10     float money, sellingPrice, expenses;
11     float goldGram, silverGram, landMarla;
12
13     float zakat;
14
15     cout << "\n===== ";
16     cout << "\n      ZAKAT CALCULATOR";
17     cout << "\n===== \n";
18
19     cout << "\nAssalam-u-Alaikum!\n";
20
21     cout << "\nEnter Your Name: ";
22     getline(cin, name);
23
24     cout << "Enter Gender: ";
25     getline(cin, gender);
26
27     cout << "Enter Age: ";
28     cin >> age;
29
30     cout << "\n-----\n";
31     cout << "Zakat is 2.5% of the wealth of each";
32     cout << "\nMuslim and is paid once a year.";
33     cout << "\n-----\n";
34
35     do
36     {

```

(globals)

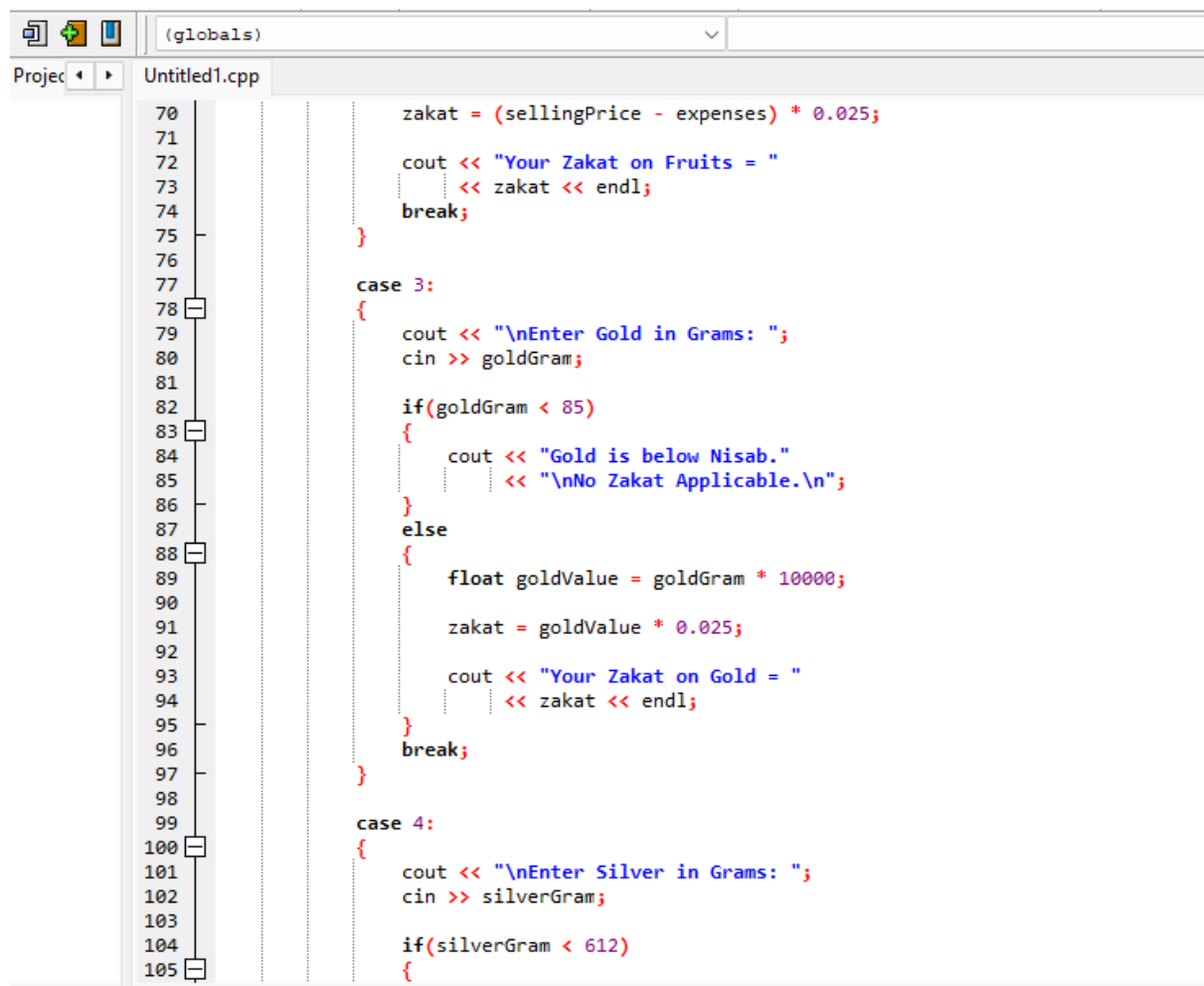
Projec | Untitled1.cpp

```

36  {
37      cout << "\n\n==== MENU =====\n";
38      cout << "\n1. Money";
39      cout << "\n2. Fruits";
40      cout << "\n3. Gold";
41      cout << "\n4. Silver";
42      cout << "\n5. Land";
43      cout << "\n6. Exit";
44
45      cout << "\n\nEnter Your Choice: ";
46      cin >> choice;
47
48      switch(choice)
49      {
50          case 1:
51          {
52              cout << "\nEnter Total Money: ";
53              cin >> money;
54
55              zakat = money * 0.025;
56
57              cout << "Your Zakat on Money = "
58                  << zakat << endl;
59              break;
60          }
61
62          case 2:
63          {
64              cout << "\nEnter Selling Price of Fruits: ";
65              cin >> sellingPrice;
66
67              cout << "Enter Expenses: ";
68              cin >> expenses;
69
70              zakat = (sellingPrice - expenses) * 0.025;
71

```

Compiler | Resources | Compile Log | Debug | Find Results



```
70      zakat = (sellingPrice - expenses) * 0.025;
71
72      cout << "Your Zakat on Fruits = "
73           << zakat << endl;
74      break;
75  }
76
77  case 3:
78  {
79      cout << "\nEnter Gold in Grams: ";
80      cin >> goldGram;
81
82      if(goldGram < 85)
83      {
84          cout << "Gold is below Nisab."
85               << "\nNo Zakat Applicable.\n";
86      }
87      else
88      {
89          float goldValue = goldGram * 10000;
90
91          zakat = goldValue * 0.025;
92
93          cout << "Your Zakat on Gold = "
94               << zakat << endl;
95      }
96      break;
97  }
98
99  case 4:
100  {
101      cout << "\nEnter Silver in Grams: ";
102      cin >> silverGram;
103
104      if(silverGram < 612)
105      {
```

OUTPUT:

```
=====
                        ZAKAT CALCULATOR
=====

Assalam-u-Alaikum!

Enter Your Name: Zuha Kamal
Enter Gender: Female
Enter Age: 21

-----
Zakat is 2.5% of the wealth of each
Muslim and is paid once a year.
-----

===== MENU =====
1. Money
2. Fruits
3. Gold
4. Silver
5. Land
6. Exit

Enter Your Choice: 1

Enter Total Money: 100000
Your Zakat on Money = 2500

===== MENU =====
1. Money
2. Fruits
3. Gold
4. Silver
5. Land
6. Exit

Enter Your Choice: 2

Enter Selling Price of Fruits: 50000
Enter Expenses: 10000
Your Zakat on Fruits = 1000
```

